Question 1 T1

Pierre pushes his car into a garage. The radius of a tire on his car is 22 cm. Determine the distance travelled by his car if the tire rotated a total of 1000°.

Solution

$$\theta = (1000)\left(\frac{\pi}{180}\right)$$
$$= \frac{50\pi}{9}$$

1 mark for conversion

$$s = \theta r$$
$$s = \left(\frac{50\pi}{9}\right)(22)$$
$$= \frac{1100\pi}{9}\text{ cm}$$

1 mark for substitution

2 marks

or

$$s = 383.972\text{ cm}$$

Question 2 R10

Solve, algebraically.

$$7^{\frac{x}{2}} = 85$$

Solution

$\log 7^{\frac{x}{2}} = \log 85$ ½ mark for applying logarithms

$\frac{x}{2}\log 7 = \log 85$ 1 mark for power law

$x\log 7 = 2\log 85$

$x = \frac{2\log 85}{\log 7}$

$x = 4.566\,142$ ½ mark for evaluating quotient of logarithms

$x = 4.566$

2 marks

Question 3 T5

Solve, algebraically, over the interval $[0, 2\pi)$.

$$\sin x(\sec x + 3) = 0$$

Solution

$\sin x = 0$	$\sec x = -3$	½ mark for solving for $\sin x$ ½ mark for solving for $\sec x$
	$\cos x = -\dfrac{1}{3}$	1 mark for reciprocal
	$x_r = 1.230\ 959$	
$x = 0,\ \pi$	$x = 1.911,\ 4.373$	2 marks for values of x (1 mark for each branch)

4 marks

Question 4 R10

Brahim invests \$2500 at an annual interest rate of 6.75% compounded monthly. Determine, algebraically, how many years it will take for his investment to reach an amount of \$10 500.

Use the formula:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

where A = the amount of the investment after t years

P = the principal of the investment

r = the annual interest rate (as a decimal)

n = the number of compounding periods per year

t = the length of the investment in years

Solution

$10\ 500 = 2500\left(1 + \frac{0.0675}{12}\right)^{12t}$	½ mark for substitution
$4.2 = (1.005\ 625)^{12t}$	½ mark for simplification
$\log 4.2 = 12t \log 1.005\ 625$	½ mark for applying logarithms 1 mark for power law
$\frac{\log 4.2}{12 \log 1.005\ 625} = t$	
$21.320\ 250 = t$	
$21.320 \text{ years} = t$	½ mark for evaluating quotient of logarithms

3 marks

There are 13 adults and 18 children who can be selected to go on a trip. Determine the number of ways 4 adults and 7 children can be selected if Sandra, one of the adults, must be selected.

Solution

${}_{1}C_{1} \cdot {}_{12}C_{3} \cdot {}_{18}C_{7}$	1 mark for ${}_{12}C_{3}$ ½ mark for ${}_{18}C_{7}$
7 001 280	½ mark for product of combinations

2 marks

Note:

- ${}_{1}C_{1}$ does not need to be shown.

In the binomial expansion of $\left(\frac{2}{x^2} - x^3\right)^9$, determine and simplify the 6th term.

Solution

$t_6 = {}_9C_5\left(\frac{2}{x^2}\right)^4\left(-x^3\right)^5$ 2 marks (1 mark for ${}_9C_5$; ½ mark for each consistent factor)

$t_6 = 126\left(\frac{16}{x^8}\right)\left(-x^{15}\right)$

$t_6 = -2016x^7$ 1 mark for simplification (½ mark for coefficient; ½ mark for exponent)

3 marks

Question 7 R1

Given that $f(x) = \{(-1, 0), (0, 2), (1, -3), (2, 4)\}$, evaluate $f(f(0))$.

Solution

$f(f(0))$

$f(2)$ ½ mark for $f(0) = 2$

4 ½ mark for $f(f(0))$

1 mark

Question 8 T2

The point $\left(-\frac{5}{6}, b\right)$ is on the unit circle and is in quadrant III.

Determine the exact value of b.

Solution

$\left(-\frac{5}{6}\right)^2 + b^2 = 1$ ½ mark for substitution

$$b^2 = 1 - \frac{25}{36}$$

$$b^2 = \frac{11}{36}$$

$$b = \pm\frac{\sqrt{11}}{6}$$

$b = -\frac{\sqrt{11}}{6}$ ½ mark for the exact value of b

1 mark

Question 9 P4

Given the following row of Pascal's Triangle, determine the values of the next row.

1 6 15 20 15 6 1

Solution

1 7 21 35 35 21 7 1

1 mark

Question 10 R3, R5

The following transformations are applied to $f(x)$, resulting in a new function, $g(x)$.

- reflection over the x-axis
- vertical stretch by a factor of 3
- horizontal stretch by a factor of 4

State the equation of $g(x)$ in terms of $f(x)$.

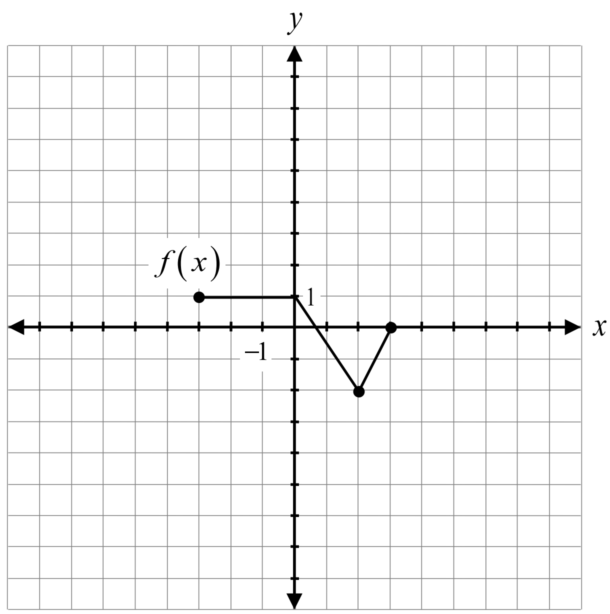
Solution

$$g(x) = -3f\left(\frac{1}{4}x\right)$$

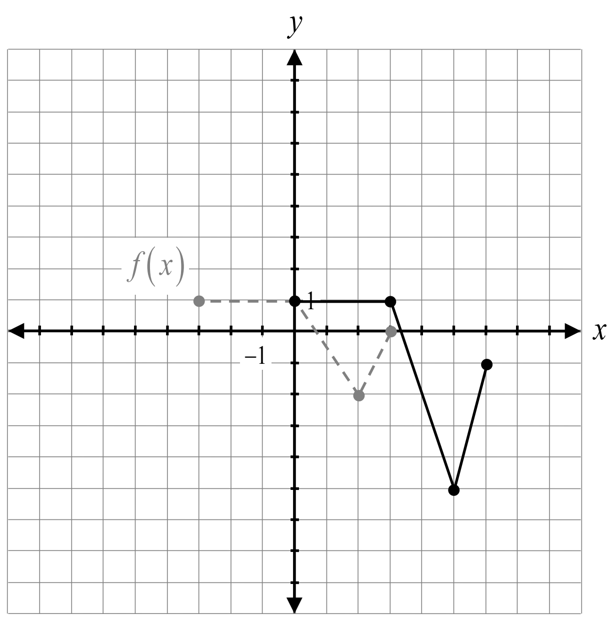
1 mark for vertical reflection
1 mark for vertical stretch
1 mark for horizontal stretch

3 marks

Given the graph of $f(x)$, sketch the graph of $y+1=2f(x-3)$.



Solution



1 mark for vertical stretch
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Question 12 R14

State the equation of the horizontal asymptote of $f(x) = \dfrac{2x^2 - 3x + 5}{4x^2 + 2x - 7}$.

Solution

$y = \dfrac{1}{2}$

1 mark

Question 13 R11

Given that $(x+4)$ is one of the factors of $p(x) = x^3 + 6x^2 - 32$, express $p(x)$ in completely factored form.

Solution

$$\begin{array}{r|rrrr} -4 & 1 & 6 & 0 & -32 \\ & \downarrow & -4 & -8 & 32 \\ \hline & 1 & 2 & -8 & 0 \end{array}$$

½ mark for $x = -4$

1 mark for synthetic division (or any equivalent strategy)

$p(x) = (x+4)(x^2 + 2x - 8)$

$p(x) = \underline{(x+4)(x+4)(x-2)}$

½ mark for consistent product of factors

or

2 marks

$p(x) = \underline{(x+4)^2(x-2)}$

Question 14 T6

Prove the identity for all permissible values of θ.

$$\frac{2\cos^2\theta}{1-\cot\theta} = \frac{\sin 2\theta}{\tan\theta - 1}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\dfrac{2\cos^2\theta}{1-\dfrac{\cos\theta}{\sin\theta}}$	$\dfrac{2\sin\theta\cos\theta}{\dfrac{\sin\theta}{\cos\theta}-1}$
$\dfrac{2\cos^2\theta}{\dfrac{\sin\theta-\cos\theta}{\sin\theta}}$	$\dfrac{2\sin\theta\cos\theta}{\dfrac{\sin\theta-\cos\theta}{\cos\theta}}$
$\dfrac{2\cos^2\theta\sin\theta}{\sin\theta-\cos\theta}$	$\dfrac{2\sin\theta\cos^2\theta}{\sin\theta-\cos\theta}$

1 mark for correct substitution of identities
1 mark for algebraic strategies
1 mark for logical process to prove the identity

3 marks

Solution

Method 2

Left-Hand Side	Right-Hand Side
$\dfrac{2\cos^2\theta}{1-\cot\theta}$	$\dfrac{2\sin\theta\cos\theta}{\dfrac{\sin\theta}{\cos\theta}-1}$
	$\dfrac{2\sin\theta\cos\theta}{\dfrac{\sin\theta-\cos\theta}{\cos\theta}}$
	$\dfrac{2\sin\theta\cos^2\theta}{\sin\theta-\cos\theta}$
	$\dfrac{\sin\theta\left(2\cos^2\theta\right)}{\sin\theta\left(1-\dfrac{\cos\theta}{\sin\theta}\right)}$
	$\dfrac{2\cos^2\theta}{1-\cot\theta}$

1 mark for correct substitution of identities
1 mark for algebraic strategies
1 mark for logical process to prove the identity

3 marks

Question 14 T6

Solution

Method 3

Left-Hand Side	Right-Hand Side
$\dfrac{2\cos^2\theta}{1-\cot\theta}$	$\dfrac{\sin 2\theta}{\tan\theta - 1}$
$\dfrac{2\cos^2\theta}{1-\dfrac{1}{\tan\theta}}$	
$\dfrac{2\cos^2\theta}{\dfrac{\tan\theta-1}{\tan\theta}}$	
$\dfrac{2\cos^2\theta}{\tan\theta-1}(\tan\theta)$	
$\dfrac{2\cos^2\theta\left(\dfrac{\sin\theta}{\cos\theta}\right)}{\tan\theta-1}$	
$\dfrac{2\cos\theta\sin\theta}{\tan\theta-1}$	
$\dfrac{\sin 2\theta}{\tan\theta-1}$	

1 mark for algebraic strategies
1 mark for logical process to prove the identity
1 mark for correct substitution of identities

3 marks

Question 15 P1

Restaurant A has 5 types of hamburgers, 2 types of french fries, and 10 types of drinks.
Restaurant B has 4 types of hamburgers, 5 types of french fries, and 6 types of drinks.

If a meal is made up of a hamburger, french fries, and a drink, justify which restaurant offers a greater variety of meals.

Solution

Restaurant A: $5 \bullet 2 \bullet 10 = 100$ meals
Restaurant B: $4 \bullet 5 \bullet 6 \quad = 120$ meals

Restaurant B offers a greater variety of meals.

1 mark

Question 16 R8

Express $\log_7(2x-5)+2\log_7 3$ as a single logarithm.

Solution

$\log_7(2x-5)+\log_7 3^2$ 1 mark for power law

$\log_7(2x-5)+\log_7 9$

$\log_7(9(2x-5))$ 1 mark for product law

or

2 marks

$\log_7(18x-45)$

Question 17 P2

Explain why 11! is not the total number of 11-letter arrangements that can be made from the word CELEBRATION.

Solution

The total number of possible arrangements is half of 11! because switching the two Es creates duplicate arrangements.

1 mark

Question 18 R13

Given $f(x) = x - 1$, identify a point on the graph of $y = \sqrt{f(x)}$.

a) $(0,-1)$

b) $(3,2)$

c) $(1,0)$

d) $(0,1)$

Question 19 P2

Identify the total number of possible arrangements of 6 adults and 4 children seated in a row if the children must sit together.

a) $6!4!$

b) $7!4!$

c) $10!$

d) $6!$

Question 20 T3

Identify the exact value of $\sec\left(-\frac{7\pi}{3}\right)$.

a) -2

b) $-\frac{2}{\sqrt{3}}$

c) $\frac{2}{\sqrt{3}}$

d) 2

Question 21 R7

Given $\log_x\left(\frac{1}{25}\right) = -2,$ identify the value of x.

a) -5

b) $-\frac{1}{5}$

c) $\frac{1}{5}$

d) 5

Question 22 T4

Identify the equation for all of the asymptotes on the graph of $y = \tan x$.

a) $x = k\pi,\ k \in \mathbb{Z}$

b) $x = 2k\pi,\ k \in \mathbb{Z}$

c) $x = \frac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$

d) $x = \frac{\pi}{2} + 2k\pi,\ k \in \mathbb{Z}$

Question 23 R12

If $p(x) = 3(m)(x+1)^2$ is a cubic function with a y-intercept of $-12,$ identify the missing factor, m.

a) $m = x - 4$

b) $m = x + 4$

c) $m = x + 12$

d) $m = x - 12$

Question 24 P4

Identify the number of negative terms in the binomial expansion of $(x-y)^5$.

a) 2

b) 3

c) 5

d) 6

Question 25 R2

Given $f(x)=x^2$, identify which equation represents the graph of $y=f(x)$ after a translation of 5 units to the left.

a) $y=(x+5)^2$

b) $y=(x-5)^2$

c) $y=x^2-5$

d) $y=x^2+5$

Question 26 R11

When a polynomial, $p(x)$, is divided by $(x-7)$, the remainder is 24. Identify the only statement that must be true.

a) $x=7$ is a zero of $p(x)$

b) $p(7)=24$

c) $x=24$ is a zero of $p(x)$

d) the y-intercept is 24

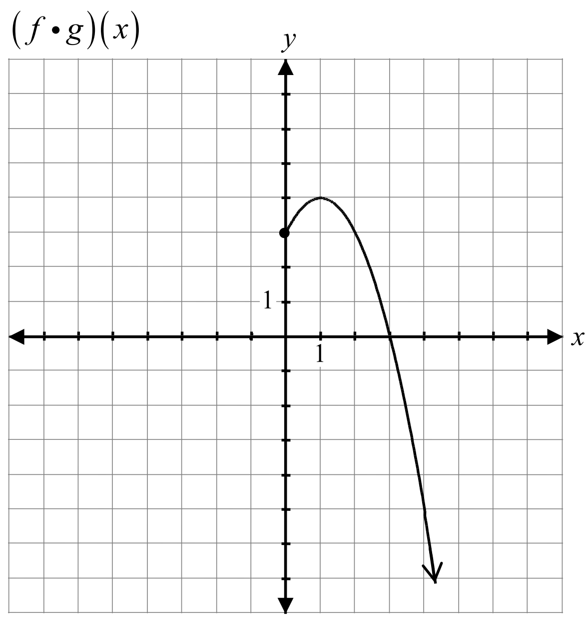
Question 27 R14

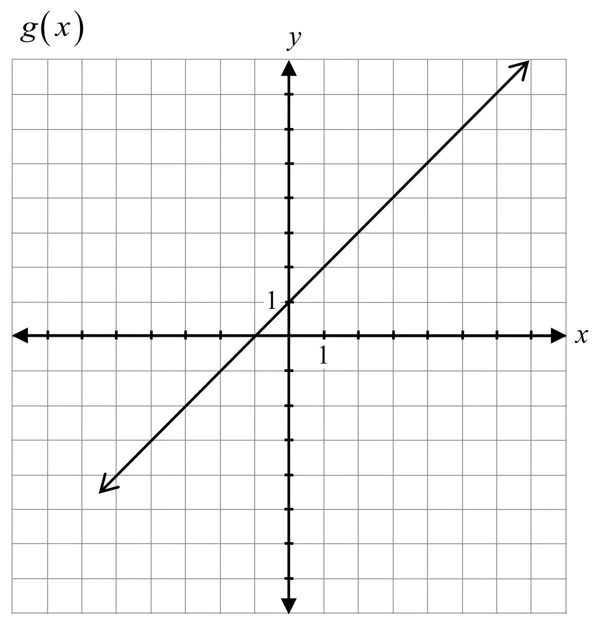
Given $f(x)=\dfrac{(2x+1)(x-8)}{(x-8)(x+4)}$, state the equation(s) of the vertical asymptote(s).

Solution

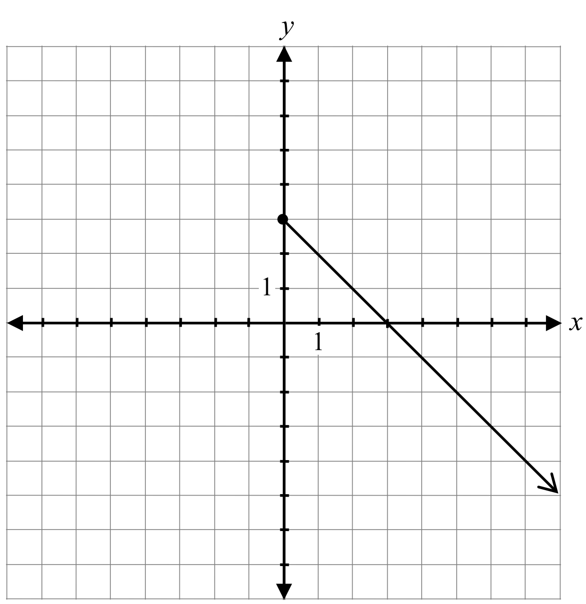
$x=-4$ **1 mark**

Given the graphs of $(f \bullet g)(x)$ and $g(x)$, sketch the graph of $f(x)$.





Solution



1 mark for operation of division
1 mark for restricted domain

2 marks

Question 29 R11

Brian was asked to state the zeros of the polynomial $p(x) = (x+2)(x-5)(x-1)$.

Brian's response:

Zeros: $(x+2), (x-5), (x-1)$

Explain why his response is incorrect.

Solution

Brian stated the factors instead of the zeros.

1 mark

Question 30 P3

Simplify ${}_{n+3}C_2$.

Solution

$$\frac{(n+3)!}{(n+3-2)!2!}$$ ½ mark for substitution

$$\frac{(n+3)!}{(n+1)!2!}$$

$$\frac{(n+3)(n+2)\cancel{(n+1)!}}{\cancel{(n+1)!}2}$$ 1 mark for factorial expansion

$$\frac{(n+3)(n+2)}{2}$$ ½ mark for simplification of factorials

2 marks

or

$$\frac{n^2+5n+6}{2}$$

Question 31 T5

Verify that the equation $2\cos^2 x = \sin x + 1$ is true for $x = \frac{\pi}{6}$.

Solution

Left-Hand Side	Right-Hand Side	
$2\cos^2\left(\frac{\pi}{6}\right)$	$\sin\left(\frac{\pi}{6}\right)+1$	½ mark for substitution
$2\left(\frac{\sqrt{3}}{2}\right)^2$	$\frac{1}{2}+1$	1 mark for exact values (½ mark for each)
$2\left(\frac{3}{4}\right)$		
$\frac{3}{2}$	$\frac{3}{2}$	½ mark for simplification

$\therefore$ LHS = RHS

2 marks

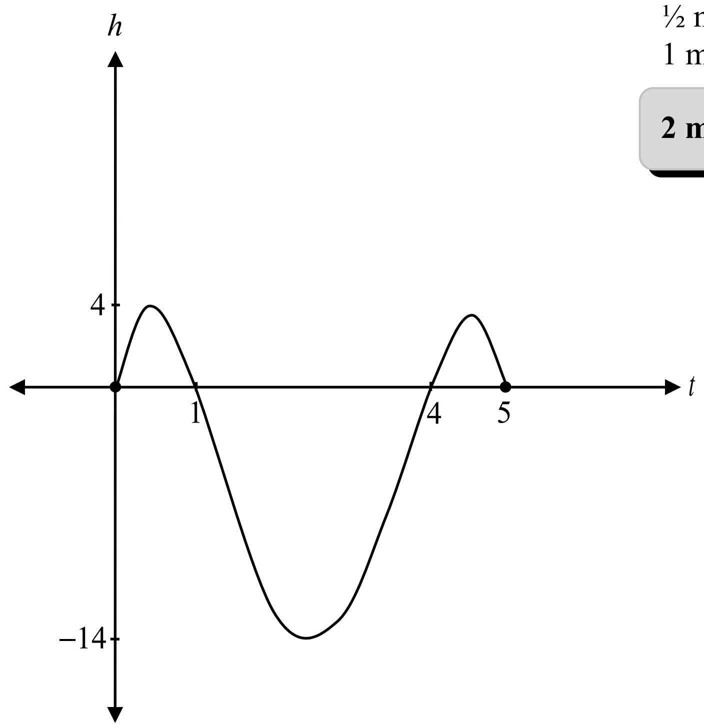
The height of a fish jumping out of the water can be modelled by the function $h(t) = -t(t-1)(t-4)(t-5)$ where $h(t)$ is the height of the fish above or below the water in cm, and t is the time in seconds, $t \geq 0$.

a) Sketch a graph representing the height of the fish with respect to time over the interval $[0,5]$.

b) State, from the graph in a), the total number of seconds that the fish is above the water.

Solution

a)



½ mark for h-intercept
½ mark for other t-intercepts
1 mark for vertical reflection of a quartic function

2 marks

b) 2 seconds

1 mark for time consistent with graph in a)

1 mark

Note:

- Scale values on h-axis are not required.

Question 33 R9

Describe the behaviour of the graph of $y = 5^x + 4$ as it approaches $y = 4$.

Solution

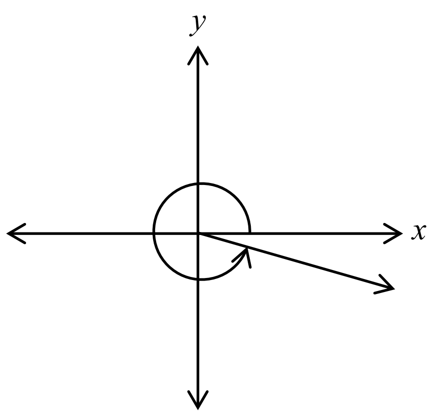
There is a horizontal asymptote at $y = 4$, so the graph approaches $y = 4$ without ever touching it.

1 mark

Question 34 T1

Sketch the angle of 6 radians in standard position.

Solution



½ mark for an angle in quadrant IV
½ mark for an appropriate angle

1 mark

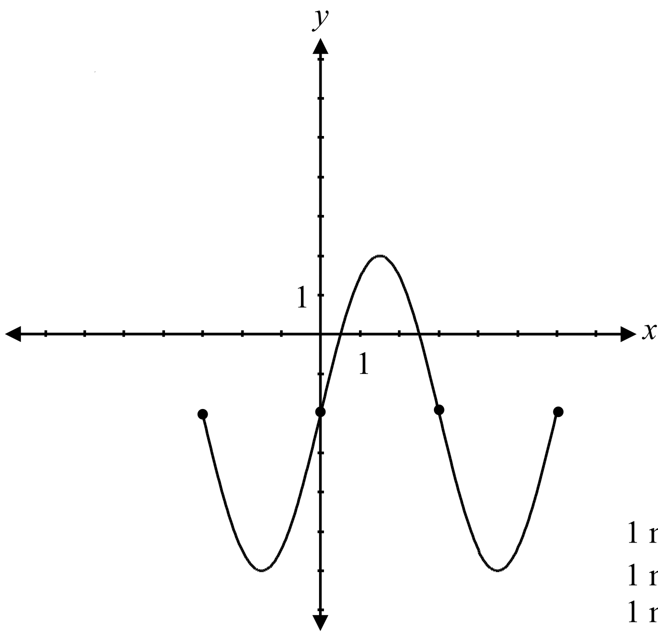
Note:

- If the directional arrow is not indicated, deduct an E1 error (final answer not stated).

Question 35 T4

Sketch the graph of the function $y = 4\sin\left(\frac{\pi}{3}x\right) - 2$ over the domain $[-3, 6]$.

Solution



1 mark for shape of $y = \sin x$
1 mark for amplitude
1 mark for period
1 mark for vertical translation

4 marks

Note:

- Deduct ½ mark for procedural error for not completing the domain of $[-3, 6]$.
- If period mark not awarded, do not deduct for E9 error (scale value on x-axis not indicated).

Question 36 R1, R14

Given $f(x) = 3x - 12$ and $g(x) = x - 4$,

a) determine the equation of $h(x) = \left(\dfrac{f}{g}\right)(x)$.

b) describe what the non-permissible value represents on the graph of $h(x)$.

Solution

a) $h(x) = \dfrac{3x - 12}{x - 4}$

1 mark for division of $\left(\dfrac{f}{g}\right)(x)$

1 mark

$$h(x) = \frac{3\cancel{(x-4)}}{\cancel{(x-4)}}$$

$$h(x) = \underline{3,\ x \neq 4}$$

b) The non-permissible value is a point of discontinuity (hole).

1 mark

Question 37 R13

Determine an equation of a radical function, $f(x)$, with a domain of $x \geq 5$ and a range of $y \geq -2$.

Solution

$f(x) = \sqrt{x-5} - 2$

1 mark for horizontal translation
1 mark for vertical translation

2 marks

Note:

- Other equations are possible.

Question 38 T6

Determine the exact value of $\cos\left(\frac{17\pi}{12}\right)$.

Solution

$$\cos\left(\frac{17\pi}{12}\right) = \cos\left(\frac{3\pi}{4} + \frac{2\pi}{3}\right)$$

$$= \cos\frac{3\pi}{4}\cos\frac{2\pi}{3} - \sin\frac{3\pi}{4}\sin\frac{2\pi}{3}$$

1 mark for substitution into correct identity

$$= \left(-\frac{\sqrt{2}}{2}\right)\left(-\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

2 marks for exact values (½ mark for each)

3 marks

$$= \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}$$

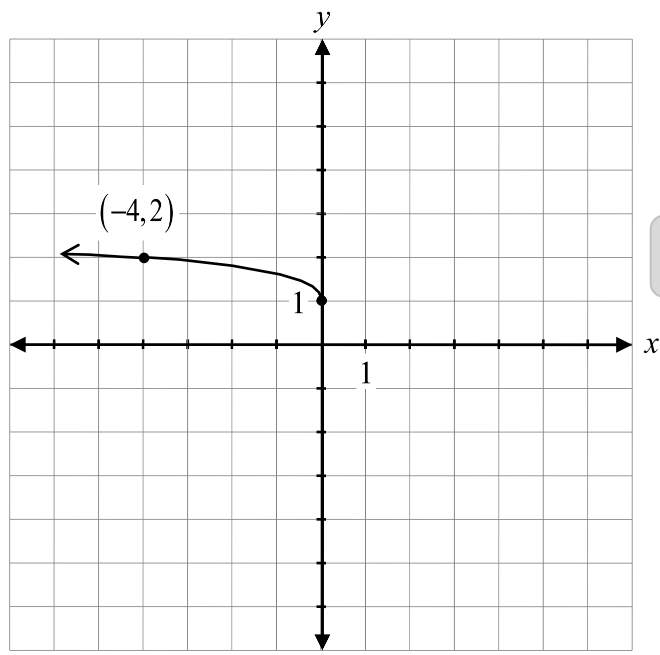
$$= \frac{\sqrt{2} - \sqrt{6}}{4}$$

Note:

- Other combinations are possible.
- Deduct a maximum ½ mark for arithmetic errors in simplification.

Sketch the graph of $y = \frac{1}{2}\sqrt{-x} + 1$.

Solution



1 mark for shape of a radical function
1 mark for vertical compression
1 mark for horizontal reflection
1 mark for vertical translation

4 marks

Question 40 R6

Given $f(x) = \frac{3x}{4} + 9$, determine the equation of $f^{-1}(x)$.

Solution

Let $y = f(x)$

$$y = \frac{3x}{4} + 9$$

$$x = \frac{3y}{4} + 9$$ 1 mark for switching x and y values

$$4x = 3y + 36$$

$$3y = 4x - 36$$

$$y = \frac{4}{3}x - 12$$ ½ mark for solving for y

$$f^{-1}(x) = \frac{4x}{3} - 12$$ ½ mark for writing equation of $f^{-1}(x)$

2 marks

Describe the end behaviour of the polynomial function $p(x) = -(x-2)(x+3)^2$.

Solution

The graph rises as x approaches negative infinity and falls as x approaches positive infinity.

1 mark

Given $\csc\theta = -4$ and θ is in quadrant IV,

a) determine the exact value of $\cos\theta$.

b) determine the exact value of $\cot\theta$.

Solution

a) $\sin\theta = -\frac{1}{4}$ 1 mark for reciprocal

$$1^2 - \left(-\frac{1}{4}\right)^2 = \cos^2\theta$$

$$\frac{15}{16} = \cos^2\theta$$

$$\pm\frac{\sqrt{15}}{4} = \cos\theta$$

$$\cos\theta = \frac{\sqrt{15}}{4}$$

1 mark for $\cos\theta$ (½ mark for quadrant; ½ mark for value)

2 marks

b)

$$\cot\theta = \frac{\frac{\sqrt{15}}{4}}{-\frac{1}{4}}$$

$$= -\sqrt{15}$$

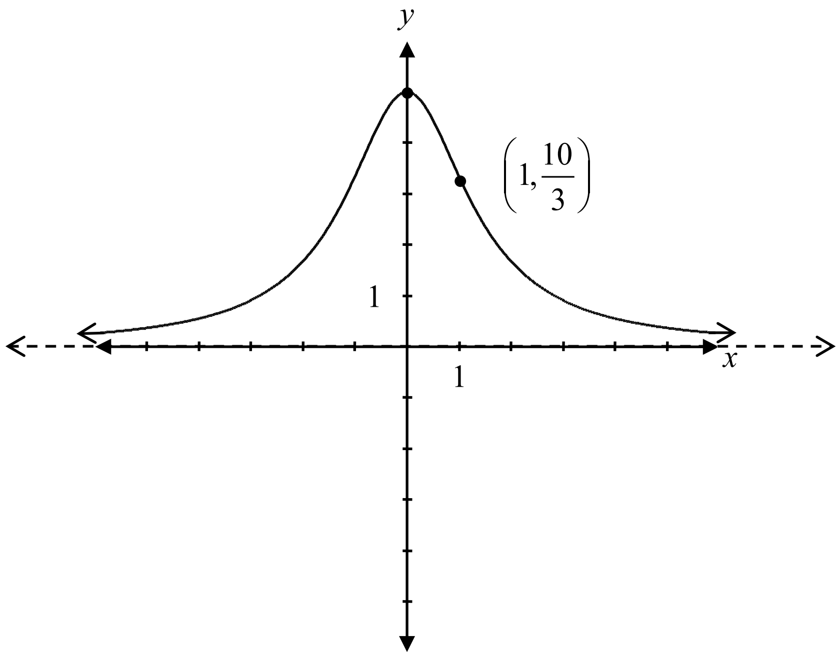
1 mark for $\cot\theta$ consistent with answer in a)
(½ mark for quadrant; ½ mark for value)

1 mark

Question 43 R14

Sketch the graph of the function $f(x)=\dfrac{10}{x^2+2}$.

Solution



1 mark for asymptotic behaviour approaching $y = 0$

1 mark for shape (½ mark for graph left of y-axis; ½ mark for graph right of y-axis)

2 marks

Question 44 R10

Explain why only one of the following equations could be solved algebraically without using logarithms.

$3^{5x} = 6^{2x-1}$ or $16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$

Solution

The equation $16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$ can be solved without the use of logarithms because 16 and $\frac{1}{2}$ can be changed to a common base of 2.

1 mark

Question 45 R1

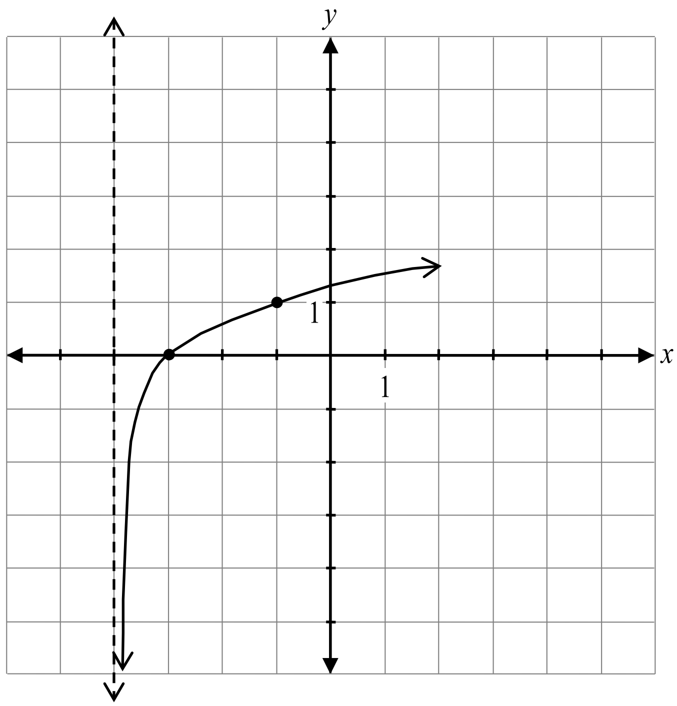
Given a graph of $y = f(x)$, describe how to sketch the graph of $y = \left|f(x)\right|$.

Solution

Reflect all the points with negative y-values over the x-axis. **1 mark**

Sketch the graph of $y = \log_3 (x + 4)$.

Solution



1 mark for increasing logarithmic function
1 mark for asymptotic behaviour approaching $x = -4$

2 marks

Solve, algebraically.

$$\log x + \log 4 - \log(x-2) = \log 5$$

Solution

Method 1

$$\log\left(\frac{4x}{x-2}\right) = \log 5$$

1 mark for product law
1 mark for quotient law

$$\frac{4x}{x-2} = 5$$

½ mark for equating arguments

$$4x = 5(x-2)$$

$$4x = 5x - 10$$

$$10 = x$$

½ mark for solving for *x*

3 marks

Method 2

$$\log x + \log 4 - \log(x-2) - \log 5 = 0$$

$$\log\left(\frac{4x}{5(x-2)}\right) = 0$$

1 mark for product law
1 mark for quotient law

$$10^0 = \frac{4x}{5(x-2)}$$

½ mark for converting to exponential form

$$5(x-2) = 4x$$

$$5x - 10 = 4x$$

$$x = 10$$

½ mark for solving for *x*

3 marks

Question 48 T5

Given $\sin\theta = \frac{1}{2}$, determine all possible values of θ over the interval $[-2\pi, 2\pi]$.

Solution

$\theta = -\frac{11\pi}{6}, -\frac{7\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$

1 mark for positive values of θ (½ mark for each)
1 mark for negative values of θ (½ mark for each)

2 marks